

Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

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(Dated: August 16, 2026)

Under mechanically constrained loading on quasi-hexagonal (qHP) C₆₀ graphullerene, measurable deviations from additive strain–dopant energetics arise because frozen elastic incompatibility introduces a bilinear strain–composition interaction that can reach magnitudes comparable to high-throughput screening margins—not because a finite-difference estimator is merely nonzero. The additive approximation in sequential workflows corresponds to neglecting the mixed second derivative $\partial^2 E/\partial\epsilon\partial\delta$ in the total-energy functional; matched four-corner audits on B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, PBE+D3 rigid protocol) show that this term is large enough to change stability rankings for size-mismatched phosphorus ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$, $\eta \approx 8\%$ at $n=4$). The four-corner difference $\mathcal{S}$ estimates that coupling on the audited grid and is not a new thermodynamic observable. Non-additivity scales with frozen local elastic incompatibility ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) rather than dopant identity alone; ionic relaxation nearly extinguishes $|\mathcal{S}|$, consistent with a dominant elastic-work contribution under constrained load rather than a partitioned claim of intrinsic electronic cross-coupling. The audit signals when sequential additive workflows should be replaced by explicit coupled evaluations on the $E(\epsilon, \delta)$ landscape.

I. INTRODUCTION

Substitutional doping and mechanical strain are widely used to tune stability and electronic structure in covalent molecular networks [1, 2]. Sequential screens often add separate strain-only and substitution-only increments, dropping the mixed derivative $\partial^2 E/\partial\epsilon\partial\delta$ [1, 3, 4]. Cluster-expansion Hamiltonians already include composition–strain cross terms [5]; what has been missing for soft fullerene networks is a matched four-corner total-energy audit under simultaneous load. Expanding the energy per atom $e(\epsilon, x) = E(\epsilon, \delta)/N_{\text{atoms}}$ about the pristine reference gives

$$e(\epsilon, x) \approx e_0 + a\epsilon + bx + c\epsilon x + \mathcal{O}(\epsilon^2, x^2, \epsilon^2 x, \epsilon x^2), \quad (1)$$

where $c = \partial^2 e/\partial\epsilon\partial x$ is the lowest-order bilinear coupling. The continuum coefficient $\gamma \equiv \partial^2 E/\partial\epsilon\partial\delta$ characterizes the intrinsic mixed response; the four-corner estimator $\mathcal{S}$ (Sec. II C) samples that response at +3% biaxial load and is a diagnostic, not a new thermodynamic observable (Supplemental Material, Sec. I).

We study quasi-hexagonal (qHP) C₆₀ graphullerene [6–8] as an elastically soft, experimentally accessible host. Substitutional B, N, and P form a donor, acceptor, and size-mismatch triad ($\sim 7\text{--}30$ pm covalent-radius excess vs. C; Table V). Prior DFT work maps strain and doping on separate grids or reports nonlinear electronic response without matched four-corner *total-energy* audits [3, 4, 7, 9–13]; a positioning table is in Supplemental Material, Table I. A substitutional defect on a pre-strained cage, together with global biaxial load, can store elastic incompatibility that linear α does not capture. Here we test whether additive totals fail under joint (ϵ, δ) load, whether the omitted term matters under mechanically constrained configurations, and whether failure tracks covalent-radius mismatch rather

than π -electron descriptors alone. The protocol is reported for one host; transferring it requires repeating the same (ϵ, δ, n) grid.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [14] with Goedecker–Teter–Hutter (GTH) pseudopotentials, the DZVP-MOLOPT-SR-GTH atom-centered basis, and the Gaussian and plane waves (GPW) scheme.

a. Exchange-correlation and dispersion. We use PBE [15] with Grimme DFT-D3 (Becke–Johnson damping) [16] for the production dataset. Hybrids or meta-GGAs (SCAN, $r^2\text{SCAN}$) can shift absolute totals by tens of meV per atom in carbon networks; production rankings are therefore framed as *within-PBE+D3* comparisons of $\mathcal{S}$ among B, N, and P under matched load. The four-corner construction [Sec. II C] subtracts totals computed with identical XC, basis, dispersion, and numerical settings, so systematic shifts largely cancel when *ranking* dopants at fixed protocol—but cancellation is not assumed without audit: on the tetramer P grid, Grimme-D3 non-additivity contributes only $\mathcal{S}_{\text{vdw}} \approx +0.33$ meV/atom (Table I), far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$ (Table II). No meta-GGA $\mathcal{S}$ values are reported (Table I; Limitations).

b. Plane-wave mesh and Brillouin-zone sampling. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (REL_CUTOFF 50 Ry; QS_EPS_DEFAULT $10^{-10} E_{\text{h}}$). Comparisons at fixed n always use the same cutoff. Production supercells contain 60–480 atoms with in-plane lattice constants up to ~ 44 Å at $n=8$; inputs use Γ -only sampling. Because $\mathcal{S}$ is a second-order dif-

ference on the *same* supercell, uniform k -point or cutoff shifts that co-drifts all four corners largely cancel in the four-corner estimator; sensitivity must be audited on $\mathcal{S}(\text{cutoff})$ and $\mathcal{S}(k)$, not on bare totals alone. A partial cutoff audit at $6 \times C_{60}$ N and +3% shifts one corner by only ~ 0.057 meV/atom between 350 and 400 Ry, implying $|\Delta\mathcal{S}| < 0.2$ meV/atom when only the mesh drifts (Table I). Dense- k ($2 \times 2 \times 1$) four-corner benchmarks on $n=1$ B/N/P at +3% are not included in the production set (Table I; Limitations); the smallest reported $|\mathcal{S}|$ values (2.3–3.0 meV/atom for N and B at $n=1$) are therefore interpreted with explicit Γ -only caution.

c. SCF convergence and four-corner precision. Unless noted otherwise, the SCF threshold is $\text{EPS_SCF} = 10^{-6} E_h$ in inner and outer orbital-transformation loops (MAX_SCF 300; OUTER_SCF MAX 30). Charged-state workflows in the Supplemental Material tighten this to $10^{-7} E_h$; selected stiff tetramer corners may use $10^{-5} E_h$ when outer SCF cycling persists at 10^{-6} (Table II). Because $\mathcal{S}$ is built from four independently converged totals [Eq. (2)], uncertainty is propagated as $\sigma_{\mathcal{S}} = 2\sigma_e$ for uncorrelated per-atom corner errors σ_e [Eq. (5); Supplemental Material, Sec. IE] (Table I). We use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as the conservative reporting band on the $n \leq 4$ production grid.

d. Spin, charge, and mechanical boundary conditions. Substitutional periodic cells are charge-neutral closed-shell Kohn–Sham calculations (no spin polarization in the production $\mathcal{S}$ grid); Hirshfeld strain-path audits on $n=1$ B/N/P use spin-polarized KS only where odd electron counts require it (Sec. IID 3). We do not apply compensating background charge or charged supercells in the $\mathcal{S}$ audit: doping is modeled as neutral substitution.

B. Strain and doping models

Biaxial strain is $\epsilon = (a - a_0)/a_0 \times 100\%$, with a (a_0) the strained (equilibrium) in-plane lattice constant; $\delta \in \{\text{B, N, P}\}$. $\mathcal{S}$ is defined in Sec. IIC; π -projected densities of states are written π -DOS. Tetramers ($4 \times C_{60}$) substitute B, N, or P at ~ 5 at.% on a fixed site map; periodic cells place one substituent per C_{60} (~ 1.67 at.%) on one production map for $n = 1$ –8. Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain) is fit on rigid PBE+D3 tetramer grids. The substitution energy $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is not a formation enthalpy (no elemental chemical potentials).

a. Rigid vs. relaxed protocols. *Rigid strain* scales lattice vectors at fixed fractional coordinates and is the production protocol unless Table II or Fig. 2(b) states otherwise. It is an upper bound on $|\mathcal{S}|$ when internal stress cannot relax (epitaxial clamp, strong adhesion, or ultrafast load), not an equilibrium free-energy difference. *Relaxed strain* uses fixed-cell BFGS at each corner before evaluating $\mathcal{S}$ (Table II; Fig. 2(b); Sec. IID 2); ionic relaxation can reverse or nearly extinguish the rigid value. Force thresholds are 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr

(~ 0.023 and ~ 0.015 eV/Å) at $\epsilon=0$ and +3%, tighter than typical 0.02 eV/Å defaults so that near-zero relaxed $|\mathcal{S}|$ is resolved.

C. Strain–dopant coupling energy

$\mathcal{S}$ is the symmetric four-corner finite-difference estimator of $\gamma \equiv \partial^2 E / \partial \epsilon \partial \delta$ at +3% load—not an order parameter and not interchangeable with γ unless higher-order remainders are negligible (Supplemental Material, Table II and Sec. I).

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (2)$$

with $\Delta E(\epsilon, \delta) = [E(\epsilon, \delta) - E(0, \text{pristine})]/N_{\text{atoms}}$ and $\delta=0$ denoting the pristine reference at the same supercell size $n \times C_{60}$. Equivalently, on corners $(0, 0)$, $(\Delta\epsilon, 0)$, $(0, \Delta\delta)$, $(\Delta\epsilon, \Delta\delta)$,

$$\mathcal{S}(\epsilon, \delta) = \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\text{atoms}}}, \quad (3)$$

which depends only on the four corner totals (reference independence).

a. Why a four-point stencil? Eqs. (2)–(3) are the minimal rectangular stencil that uses four DFT totals, is second-order accurate in the bilinear term, cancels first-order increments and a shared offset, and equals the additive-screening bias by construction. Asymmetric three-point schemes retain linear contamination; other estimators are compared in Supplemental Material, Sec. IC.

b. Finite-difference link to the mixed derivative. Expanding each corner total to second order and forming Eq. (3) cancels $E_{\epsilon} \Delta\epsilon$ and $E_{\delta} \Delta\delta$ identically (Supplemental Material, Sec. IB), yielding

$$\mathcal{S}(\epsilon, \delta) \approx \frac{\partial^2 E}{\partial \epsilon \partial \delta} \frac{\Delta\epsilon \Delta\delta}{N_{\text{atoms}}} + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2\delta, \epsilon\delta^2), \quad (4)$$

so $\mathcal{S}$ reports the discrete mixed second derivative of total energy on the audited (ϵ, δ) grid. At finite +3% biaxial load and discrete B/N/P labels, higher-order terms in Eq. (4) are not separately resolved on the production periodic map; multi-amplitude strain audits ($\epsilon = 1$ –3%) are targeted sensitivity tests (Limitations). Throughout the main text, “nonlinear coupling” refers to a nonzero mixed response estimated by $\mathcal{S}$, not to $\mathcal{S}$ itself as a fundamental constant.

$\mathcal{S}$ is reported in meV per atom and is invariant under a uniform energy offset shared by all four corners (Supplemental Material, Sec. ID).

c. Error propagation. If each corner carries uncorrelated equal per-atom uncertainty σ_e ,

$$\sigma_{\mathcal{S}}^2 = 4\sigma_e^2, \quad \sigma_{\mathcal{S}} = 2\sigma_e, \quad (5)$$

from the four-term difference in Eq. (3) (derivation: Supplemental Material, Sec. IE). Table I lists contributions

to the reporting band; we use $\sigma_S \lesssim 2$ meV/atom as a conservative protocol floor on the $n \leq 4$ grid. $\mathcal{S} < 0$ (> 0) means the coupled corner is more (less) stable than the additive estimate. Additive-failure magnitude is

$$\eta \equiv \frac{|\mathcal{S}|}{|\Delta E_{\text{coupled}}|}, \quad (6)$$

where $\Delta E_{\text{coupled}} = (E(\epsilon, \delta) - E(0, 0))/N_{\text{atoms}}$.

D. Validation benchmarks

Interpretation rests on four checks: numerical hygiene (Table I); ionic relaxation of tetramer P and periodic $n=1$ P (Table II; Fig. 2(b)); matched-functional rigid vs. relaxed tetramer P (Table II); and alternate placement (Table III). Periodic $n=4$ P relaxation, B/N periodic relaxations, dense- k / meta-GGA four-corner checks, and two-body image tests beyond $n=8$ are not reported (Table I; Limitations). A calculation-group inventory is given only in the Supplemental Material.

1. Alternate-placement benchmark

An independent dopant configuration at the same nominal ~ 5 at.% loading repeats the B/N/P strain grid under rigid PBE+D3 (Table III). Main-text quantitative α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the reference-placement map in Fig. 3(a,b); the alternate seed 137 grid is archived only. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (2) audit; the $|\mathcal{S}|$ ranking at +3% matches the production map in Table IV.

2. Ionic-relaxation benchmark

Fixed-cell BFGS geometry optimization is performed on the tetramer pristine/ P corners at $\epsilon=0$ and +3% (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, ~ 0.023 and ~ 0.015 eV/Å; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr). These thresholds are tighter than typical 0.02 eV/Å defaults because relaxation can drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale. A matched-functional rigid PBE+D3 control at the relaxed coordinates quantifies how $\mathcal{S}$ changes upon relaxation (Table II). The same BFGS thresholds are used for the periodic $n=1$ P four-corner relaxation reported in Fig. 2(b).

3. Hirshfeld population along strain

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic $n=1$ B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table V.

TABLE I. Error budget for four-corner $\mathcal{S}$ [Eq. (3)]. $\sigma_{\mathcal{S}} \approx 2\sigma_e$ when per-atom corner errors σ_e are uncorrelated and equal [Eq. (5)]; shared XC, k , and cell settings induce partial cancellation. Rows marked “not reported” are targeted sensitivity tests (Limitations); no meta-GGA or dense- k numbers appear without converged outputs. The ± 2 meV/atom band is a conservative protocol floor, not a quadrature sum.

Source	$ \Delta\mathcal{S} $ or $\sigma_{\mathcal{S}}$ (meV/atom)	Status
SCF (EPS_SCF = 10^{-6})	$\lesssim 2$ ($\sigma_{\mathcal{S}} \approx 2\sigma_e$)	Analytic bound
Cutoff (350 vs 400 Ry)	< 0.2	Partial ($6 \times \text{C}_{60}$ N, +3%)
D3 cross term	$ \mathcal{S}_{\text{vdW}} \approx 0.33$	Tetramer P
D3 (B/N periodic)	—	Not reported
Ionic relaxation (rigid vs. relaxed P)	$\gg 2$	Table II
k mesh (Γ vs $2 \times 2 \times 1$)	—	Not reported
XC (PBE+D3 vs SCAN / r ² SCAN)	—	Not reported
<i>Adopted reporting band ($n \leq 4$): ± 2 meV/atom.</i>		

TABLE II. Ionic-relaxation benchmark for the tetramer P -substitution four-corner grid (fixed-cell geometry optimization: pristine and P at $\epsilon=0$ and +3%, same biaxial cell). Relaxed corners and the matched-functional ratio use PBE+D3 ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; sign not preserved). $\mathcal{S}$ follows Eq. (2) with $N=240$ atoms.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_{\text{h}})$	—	−1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_{\text{h}})$	—	−1363.9550
$E(P, \epsilon=0) (E_{\text{h}})$	—	−1374.4415
$E(P, \epsilon=3\%) (E_{\text{h}})$	—	−1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	−2.28
Sign of $\mathcal{S}$ vs. rigid	+	− (not preserved)

TABLE III. Archived alternate-placement tetramer grid (PBE+D3 rigid; seed 137). Not used for main-text figures or quantitative claims; retained for reproducibility of the placement-sensitivity audit. Reference-map α and $\mathcal{S}_{+3\%}^{\text{tet}}$ appear in Fig. 3(a,b). Periodic $\mathcal{S}(n)$ is in Table IV.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)
B	alternate	+21.4	−2.6
N	alternate	+45.4	+12.6
P	alternate	+989.6	+41.4

III. RESULTS

Total-energy audits use matched periodic supercells and the reference tetramer map (Sec. II C). Mechanism is deferred to Sec. IV.

A. Electronic structure under strain and doping

Figure 1(a) shows the reference tetramer: pristine and B/N/P-substituted cages at $\epsilon=0$ with highlighted substituent sites (covalent-radius excess in Table V). Panels (b)–(d) are the one-parameter slices used in sequential screens: the N π -DOS waterfall shifts with biaxial strain; $E_g(\epsilon)$ places B ($E_g \approx 0.03$ eV), N ($E_g \approx -0.14$ eV), and P ($E_g \approx 0.05$ eV) at $\epsilon=0$ on distinct trajectories while pristine remains open; panel (d) is the pristine mechanical $\Delta E(\epsilon)$ path. Supplemental Material gives full π -DOS stacks and periodic VBM/CBM isosurfaces (N size series; P $n=4$): frontier density remains cage-localized for $n = 2$ –6.

B. Non-additive strain–doping coupling

Figure 2 is the central audit. Panel (c) defines the test: strain-only and substitution-only increments are summed and compared with the coupled four-corner total; $\mathcal{S}$ [Eq. (3)] is that signed gap. Panel (a) and Table IV report the fifteen periodic +3% $\mathcal{S}(n, \delta)$ values. $|\mathcal{S}|$ is not a monotonic $1/n$ dilution: P remains large and negative through $n=8$ (2.2–31.9 meV/atom), while N changes sign at $n=8$. At $n=4$, $\mathcal{S} = -6.2 \pm 2$ (B), $+3.0 \pm 2$ (N), and -23.7 ± 2 meV/atom (P; $\eta \approx 8\%$), where $\pm$ is the conservative protocol floor (Table I). The P extremum at $n=1$ (31.9 ± 2 meV/atom) may include enhanced periodic-image coupling at the smallest cell; it is not a separate materials constant. We do not extrapolate $\mathcal{S}_\infty$. Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1. Negative $\mathcal{S}$ (B, P at $n=4$) means the coupled corner lies below the additive estimate; positive $\mathcal{S}$ (N) means it lies above. Panel (b) is the protocol bound on a different system from the tetramer row of Table II: periodic $n=1$ P at +3% falls from $\mathcal{S}_{\text{rig}} \approx -31.9$ meV/atom to near zero after ionic relaxation. The matched-functional tetramer P grid in Table II instead reverses sign (+1.19 vs. -2.28 meV/atom). Both checks show that rigid $|\mathcal{S}|$ is an upper bound, not an equilibrium free-energy difference. Panel (d) is a schematic of the bilinear $c\epsilon\delta$ term, not a DFT fit.

C. Mechanical strain sensitivity

Pristine tetramers yield $|\alpha_{\text{pris}}| \approx 1$ meV/% on the six-point reference grid. Least-squares slopes for B, N, and P are +58.7, -297.5 , and $+456.0$ meV/% [Fig. 3(d)];

horizontal bars are linear-fit standard errors]. Table V reports $\bar{d}$, $\sigma(\bar{d})$, and Δr_{cov} . At +3%, P shows $\Delta \bar{d} \approx 3 \times 10^{-5}$ Å versus $\sim 8.5 \times 10^{-3}$ Å for B.

Figure 3 traces microscopic strain paths on the reference map. Hirshfeld $q(\epsilon)$ and $\bar{d}(\epsilon)$ [panels (a,b)] vary with dopant and strain. Local disorder $\sigma(\bar{d})(\epsilon)$ [panel (c)] broadens for B and N under tension while remaining near the $\epsilon=0$ baseline for P. Panel (d) maps reference-map $|\alpha|$ against periodic $|\mathcal{S}|_{n=4}$ (marker area $\propto |\mathcal{S}|_{n=1}$): N lies in the high- $|\alpha|$ /low- $|\mathcal{S}|_{n=4}$ quadrant and P in the high- $|\mathcal{S}|$ region—a qualitative rank inversion among three dopants, not a fitted correlation. P has the largest covalent-radius mismatch ($\Delta r_{\text{cov}} = +30$ pm; Table V) and the largest audited $|\mathcal{S}|$ under the rigid protocol.

IV. DISCUSSION

A. Main finding

Additive screening fails under constrained joint load because frozen elastic incompatibility generates a bilinear $c\epsilon x$ term [Eq. (1)] large enough to change rankings for P (Table IV), while linear α does not rank $|\mathcal{S}|$ [Fig. 3(d)]. Ionic relaxation nearly extinguishes $|\mathcal{S}|$ (Table II; Fig. 2(b)), consistent with a dominant elastic-work contribution rather than a partitioned electronic claim (Sec. IV E). On this PBE+D3 map we treat $|\mathcal{S}| > \sim 2$ meV/atom or $\eta > \sim 1\%$ as a practical flag for coupled re-evaluation, not a thermodynamic boundary.

B. Mechanism

For size-mismatched P, the dominant chain is (Fig. 4): $\Delta r_{\text{cov}} = +30$ pm $\rightarrow$ frozen misfit stress $\rightarrow$ overlap with biaxial tension $\rightarrow$ bilinear elastic work $\rightarrow$ largest $|\mathcal{S}|$. *Primary* control is geometric misfit; *secondary* is electronic redistribution (Hirshfeld Δq on P $\sim 11\%$ fractional $|\Delta q/q|$, below N's $\sim 18\%$); *tertiary* is π -DOS morphology. Electronic structure responds rather than controls: π -DOS does not, by itself, rank periodic $|\mathcal{S}|$ at $n=4$ [Figs. 1(b,c) and 3(d)]. $|\mathcal{S}|$ rises with $|\Delta r_{\text{cov}}|$ for B and P at $n=1$; N breaks a radius-only trend (Fig. 4(d)). B shows the largest $\Delta \bar{d}$ at +3% yet modest $|\mathcal{S}|$; P the opposite (Table V). Ionic relaxation releases the misfit (Fig. 2(b)); Grimme D3 contributes only $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom at tetramer P (Table I).

C. Relation to literature

This is a periodic total-energy audit of joint (ϵ, δ) load (Table IV), not a strain-only, doping-only, or mobility/gap map [3, 4, 7, 9, 10]. Cluster expansions already encode $c\epsilon x$ [5]; the increment is the matched four-corner audit on a soft, pre-strained network with rigid-vs.-relaxed and placement checks (Tables II–IV; Supple-

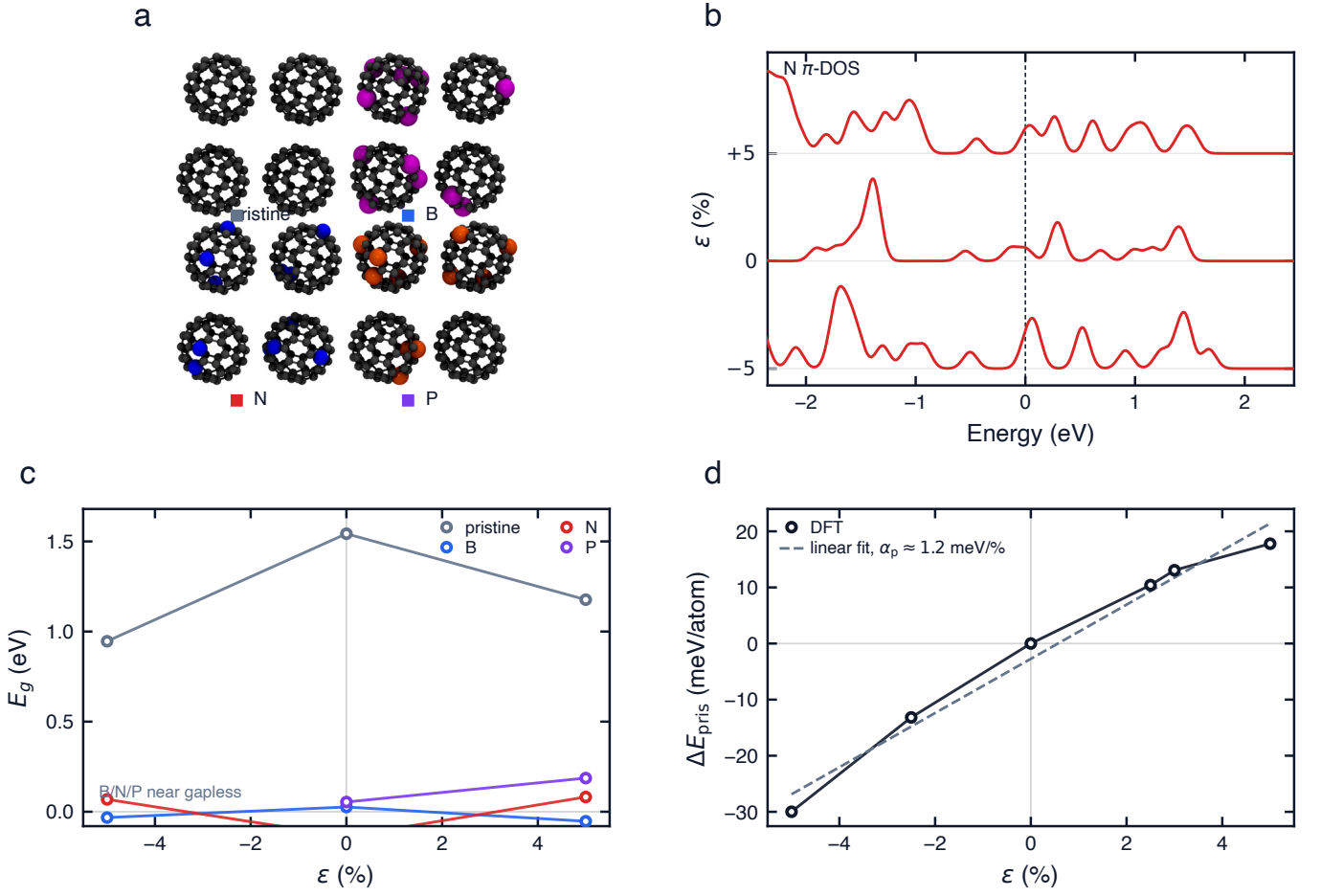


FIG. 1. Reference tetramer and one-parameter strain slices (PBE+D3). (a) qHP C₆₀ cages at $\epsilon=0$. Heteroatoms are CPK; B/N/P labels follow the manuscript palette of (c) (Table V). (b) N π -DOS vs. energy, stacked by biaxial strain $E_F=0$. (c) Gap $E_g(\epsilon)$; B/N/P remain near gapless on this scale. (d) Pristine mechanical $\Delta E(\epsilon)$ and linear α . Joint-load four-corner audits are in Fig. 2.

TABLE IV. Periodic coupling-energy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2); conservative protocol uncertainty ± 2 meV/atom for $n \leq 4$ (Sec. II C). Substitutional loading is one dopant per C₆₀ unit (~ 1.67 at.% of total atoms in each supercell). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C ₆₀ units)	Dopant	ϵ (%)	at.%	$\mathcal{S}$ (meV/atom)
1	B	+3	1.67	-8.38
1	N	+3	1.67	+2.30
1	P	+3	1.67	-31.91
2	B	+3	1.67	-2.18
2	N	+3	1.67	+6.85
2	P	+3	1.67	-28.29
4	B	+3	1.67	-6.23
4	N	+3	1.67	+2.95
4	P	+3	1.67	-23.71

mental Material, Table I). Prior graphullerene studies often relaxed ions first or added separate grids [4, 7, 9], which masks constrained-load coupling.

D. Applicability and generality

The four-corner audit applies when local substitutional strain cannot fully relax on a short-range, elastically soft covalent network [8, 17]. It does not apply to fully re-

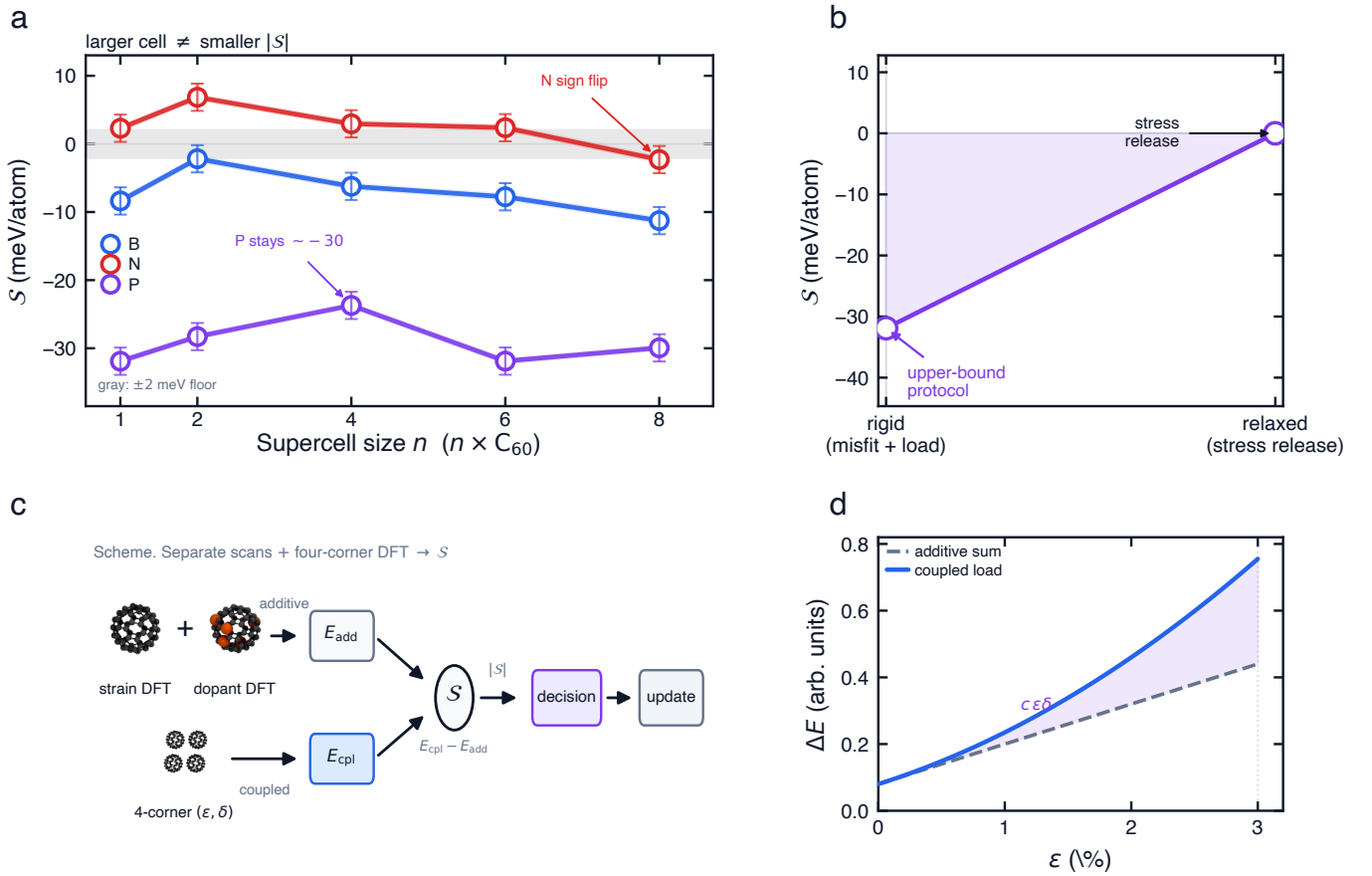


FIG. 2. Additive strain-dopant screening fails under constrained load. (a) Fifteen periodic +3% $S(n, \delta)$ values vs. supercell size (markers: DFT; error bars and gray band: ± 2 meV/atom protocol floor; Tables I and IV). $|S|$ does not fall as $1/n$: P remains ~ -24 to -32 meV/atom through $n=8$; N changes sign at $n=8$. No S_∞ extrapolation. (b) Protocol bound, not a second size-scaling panel: periodic $n=1$ P at +3%, rigid vs. ionically relaxed S (near zero after stress release). The tetramer sign reversal in Table II is a different system. (c) Definition: additive sum of strain-only and substitution-only increments vs. the coupled four-corner total; S is that gap [Eq. (3)]. (d) Schematic $E(\epsilon)$ at fixed composition: the bilinear $c\epsilon\delta$ term bends the coupled path away from the additive sum (cartoon, not a DFT fit).

TABLE V. Local dopant-cage structure from archived *tetramer XYZ* ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ tracks the uniform nearest-neighbor stretch component. Distances are DFT-derived from archived tetramer XYZ; no experimental bond errors are quoted—comparison is internal across dopants and with qHP literature on intrinsic cage strain [8, 11].

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon=+3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	-6
P	1.384	1.385	+0.00003	0.0004	+30

laxed bulk references, metallic or liquid hosts, or high- T regimes where configurational entropy dominates sub-meV differences (Sec. IVE). Uniaxial load should preserve the stencil but rescale c through elastic anisotropy; that case is not audited.

Laboratory tests need not resolve S : cage-mode Raman/IR or substrate-bending that deviate from separate strain-plus-doping calibrations would signal additive fail-

ure, provided chemistry and strain are matched and the comparison is to a constrained, not fully relaxed, state.

E. Limitations

Main-text α and S use PBE+D3 under the rigid-strain protocol; ionic relaxation can reverse or nearly extinguish

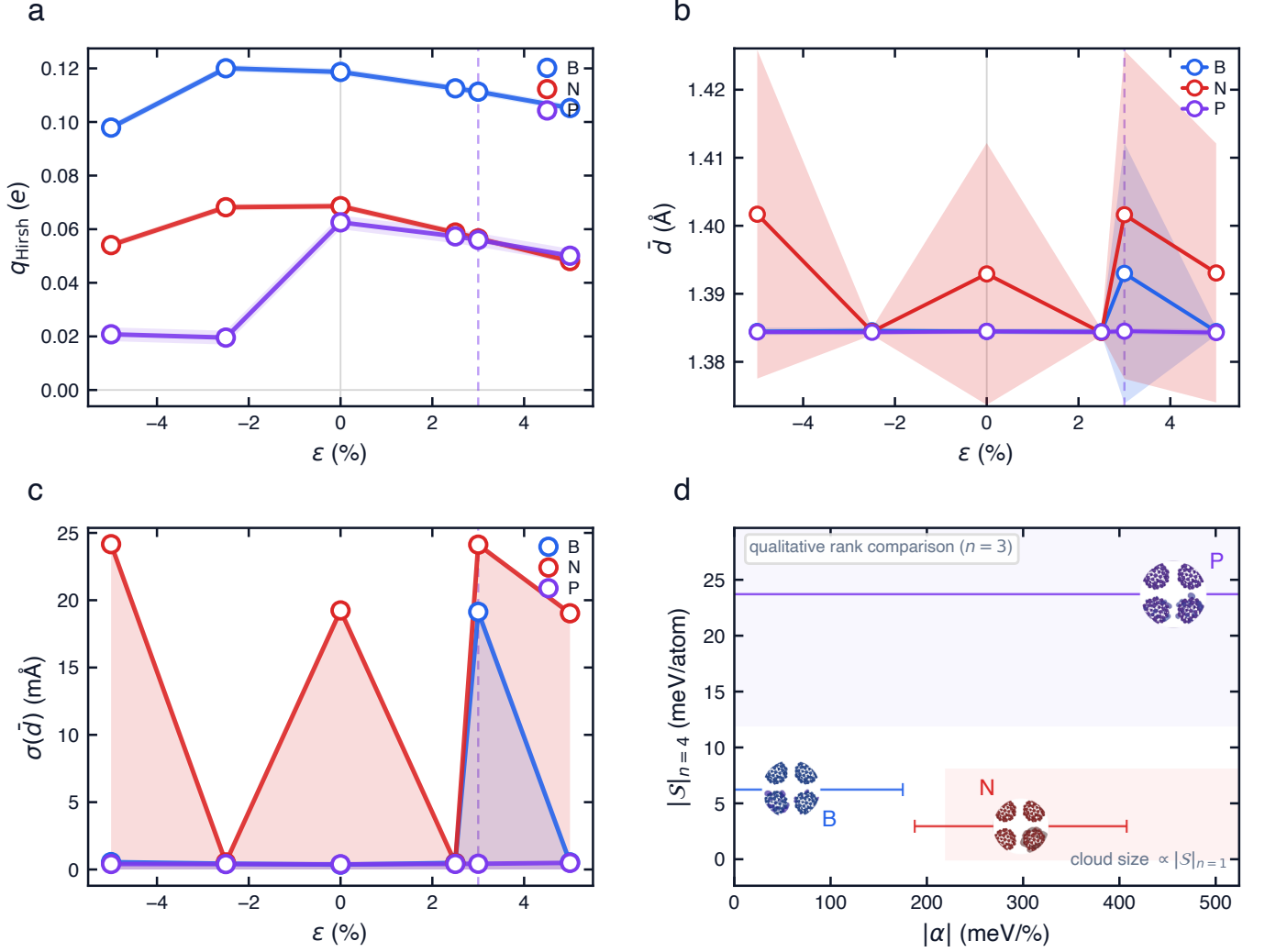


FIG. 3. Microscopic strain paths and α - $|S|$ rank comparison ($n=3$ dopants). (a) Hirshfeld $q(\epsilon)$ for B/N/P (six-point paths; dotted guide at +3%). (b) Nearest-neighbor $\bar{d}(\epsilon)$ with $\pm\sigma$ bands. (c) Local disorder path $\sigma(\bar{d})(\epsilon)$. (d) Reference-map $|\alpha|$ vs. periodic $|S|_{n=4}$ (marker area $\propto |S|_{n=1}$; horizontal bars: linear-fit standard errors on the six-point strain grid). N inverts a naive $|\alpha| \rightarrow |S|$ expectation; with only three dopants this panel reports qualitative ranking, not Pearson, R^2 , or Spearman statistics.

S (Table II). Eq. (1) and the four-corner link [Eq. (4)] assume small perturbations on a single smooth energy branch; they lose validity at large strain (e.g. $\gtrsim 10\%$) or across structural phase transitions, and the production audits intentionally remain at +3% biaxial load inside the nominally elastic regime. All totals are 0 K Born–Oppenheimer energies; finite-temperature free energies and configurational entropy remain unexplored and could reorder sub-meV competitors without restoring additivity if bilinear coupling persists at the electronic-structure level. Production periodic cells use Γ -only sampling; dense- k and SCAN/r²SCAN four-corner checks are not reported (Table I). Alternate placements (Table III) bound configuration specificity; Table V geometry is a tetramer proxy (~ 5 at.%) without Mayer/Bader partitioning or matched periodic $n=4$ bond statistics. With only three substitutional labels, the α - $|S|$ map [Fig. 3(d)]

is a qualitative rank comparison among $n=3$ chemistries, not a fitted correlation law or predictive model for linear α . Direct elastic-vs.-electronic energetic decomposition (Mayer bond order, Bader strain partitioning, virial stress) and dispersion cross-term audits for B/N periodic cells remain open (Table I).

F. Constrained load and outlook

Rigid $|S|$ bounds epitaxial clamp, adhesion, or ultra-fast load where ions cannot relax (Fig. 2(b)); fully relaxed screens are expected to recover near-additive totals and must not inherit the rigid magnitudes as free-energy differences. The tetramer P grid (Table II) is a different system and reverses sign rather than collapsing to zero. For a new host: fix or relax coordinates

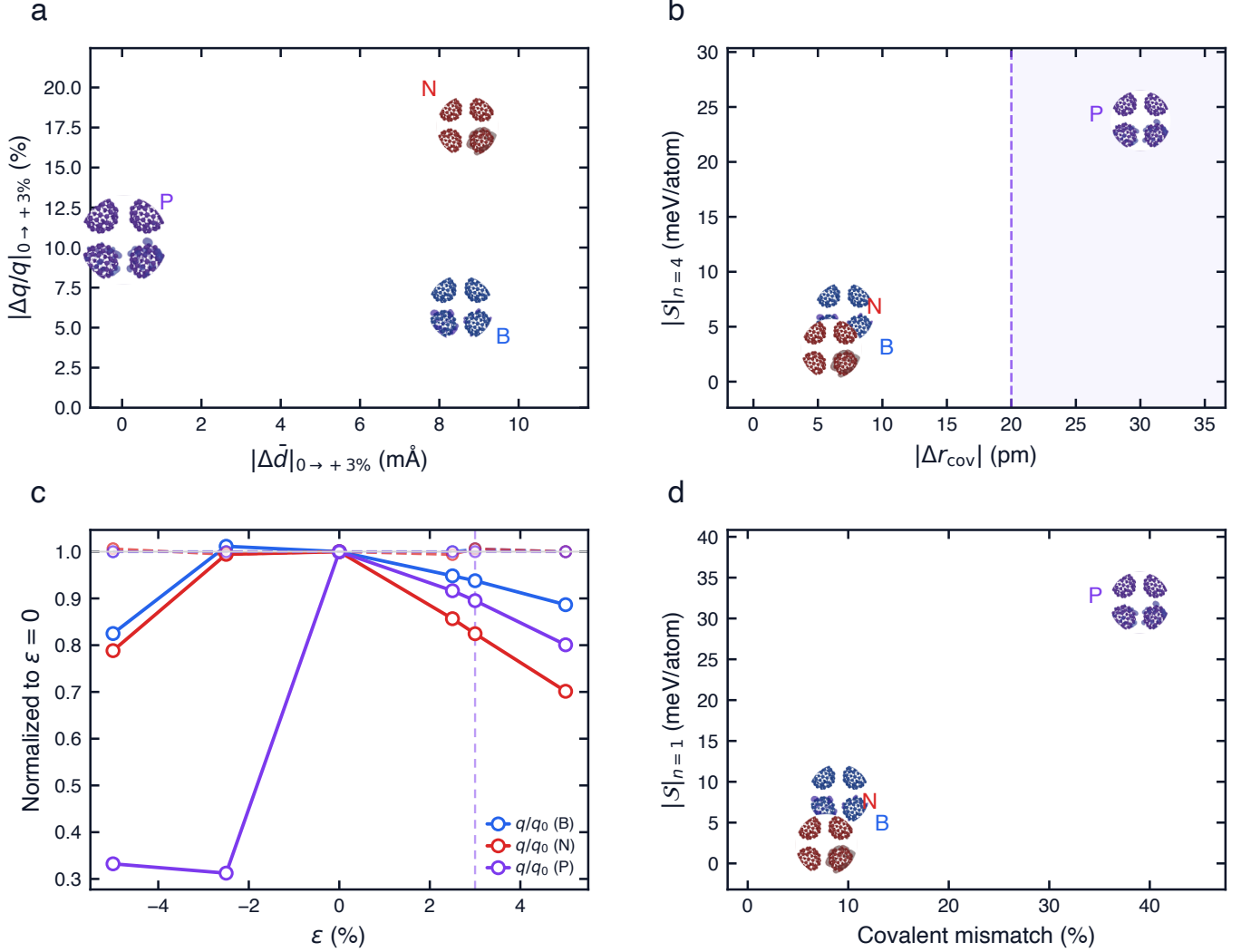


FIG. 4. Mechanism hierarchy for P-dominated coupling (primary: misfit strain $\rightarrow$ frozen internal stress; secondary: charge redistribution; tertiary: π -DOS morphology). (a) Response plane from 0 $\rightarrow$ +3% path endpoints; bubble area $\propto |S|_{n=4}$ (B: large $\Delta \bar{d}$, small $|S|$; N: large $|\Delta q|$, moderate $|S|$). (b) $|\Delta r_{\text{cov}}|$ vs. $|S|_{n=4}$: radius alone does not rank B/N/P. (c) Linked normalized paths $q(\epsilon)/q(0)$ (solid) and $\bar{d}(\epsilon)/\bar{d}(0)$ (dashed). (d) Covalent mismatch (%) vs. $|S|_{n=1}$ (P extremum at largest mismatch).

to the loading scenario, evaluate four-corner $\mathcal{S}$, and replace sequential screening if $|\mathcal{S}| > \sim 2$ meV/atom or $\eta > \sim 1\%$ (Fig. 2(c,d)). Larger-radius group-V substituents (As, Sb) are expected to couple more strongly than P under constrained load; that extrapolation is unverified. Follow-up: Mayer/Bader partitioning, $\mathcal{S}(\text{cutoff})$ and $\mathcal{S}(k)$, and periodic $n=4$ P relaxation.

V. CONCLUSIONS

Additive strain-dopant screening fails under constrained load on qHP C₆₀ because stored elastic incompatibility generates a bilinear correction large enough to reorder P rankings ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$; $\eta \approx 8\%$ at $n=4$). $\mathcal{S}$ estimates that mixed derivative; it

is not a new thermodynamic observable. Linear α does not rank $|\mathcal{S}|$ among B/N/P [Fig. 3(d)]. For P, geometric misfit is primary and ionic relaxation nearly extinguishes $|\mathcal{S}|$ (Fig. 2(b)). Coupled (ϵ, δ) evaluation is advisable for $|\Delta r_{\text{cov}}| \gtrsim 20$ pm substituents on this host under constrained +3% load; other networks require their own four-corner audit.

ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including computational inputs and outputs, tabulated coupling-energy and polaron analyses, four-corner total-

energy components underlying $\mathcal{S}$ (machine-readable synergy tables), and materials to reproduce all figures and tables in the main text. Ionic-relaxation, Hirshfeld, and size-scaling archives referenced in the main-text tables are included in the public release. A tagged release matching the manuscript revision will be issued upon acceptance.

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